

5/20/2024

Informatics 3 Year Project

Deliverable 4: Project Management

CYBER CREW



TEAM NAME: CYBER CREW

TEAM NUMBER: 4

PROJECT NAME: AUTOWORX BARNYARD ONLINE

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1. Updated Problem Statement, Proposed Solution & Use Cases

1.1 Problem Statement

AutoWorx Barnyard's reliance on outdated methods like telephone and WhatsApp for communication, Microsoft Word for invoices and quotations, and verbal task assignments results in inefficient customer relationship management, task delegation, and process documentation, hindering productivity and customer engagement.

1.2 Proposed Solution

To address the current shortcomings, we will develop a comprehensive workshop management system designed to enhance customer engagement, streamline communication, and optimize service estimation processes. This system will introduce a centralized platform for managing all aspects of the workshop's operations, thereby improving overall productivity and service quality.

Key features will include:

1. **Enhanced Customer Engagement:** A user-friendly customer portal where clients can easily book appointments, track service progress, and communicate with the workshop in real-time.
2. **Seamless Communication:** Integrated communication tools that replace telephone and WhatsApp, allowing for more structured and documented interactions with customers and within the team.
3. **Efficient Service Estimation and Invoicing:** Automated service estimation, quotation and invoice generation processes that reduce manual errors, save time, and provide customers with clear, professional documents quickly.
4. **Task Delegation and Coordination:** A task management system that assigns tasks to mechanics systematically, ensuring accountability and tracking progress in real-time, thereby reducing reliance on verbal instructions.
5. **Comprehensive Client Service History:** A database that maintains detailed records of each client's service history, enabling personalized service, better follow-ups, and improved customer satisfaction.
6. **Skill Sharing and Knowledge Management:** Tools to promote the sharing of skills and best practices among team members, ensuring consistent service quality and facilitating continuous learning and improvement.

By implementing this workshop management system, AutoWorx Barnyard will achieve greater operational efficiency, better customer service, and enhanced team collaboration, ultimately leading to higher customer satisfaction and business growth.

1.3 Use Cases

1.3.1 User Management

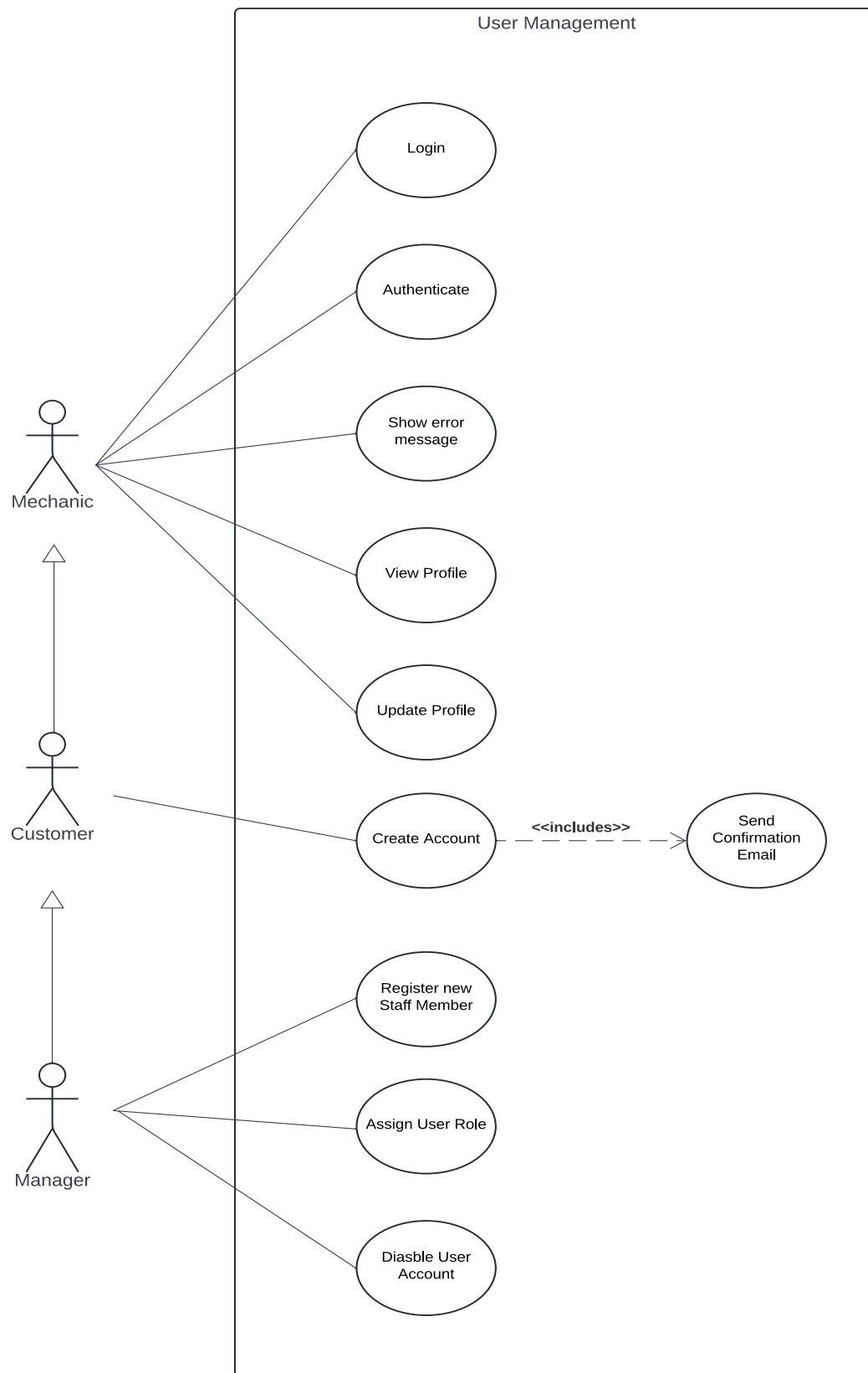


Figure 1: User Management Use Case Diagram

1.3.2 Service Request Management

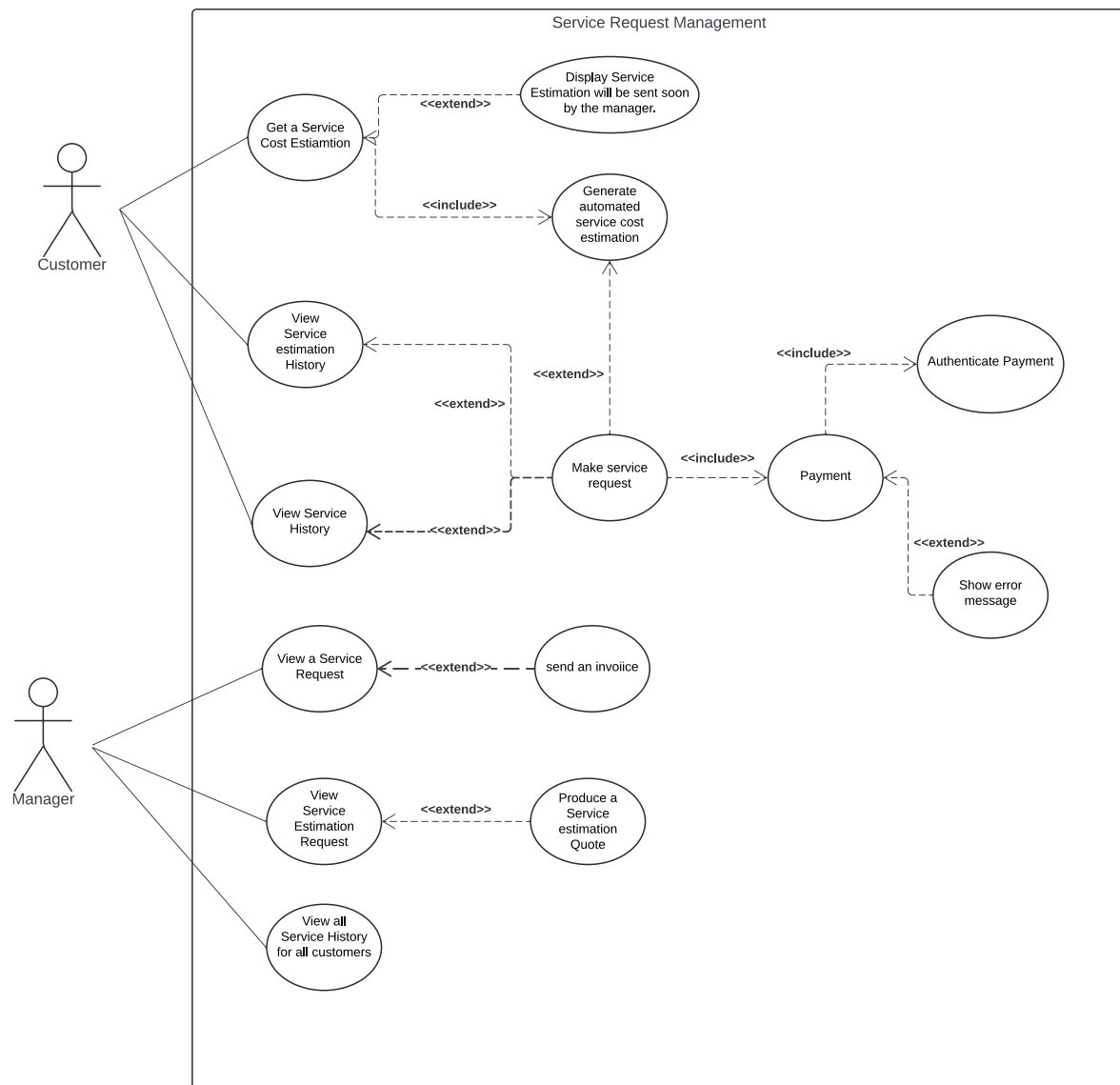


Figure 2: Service Request Management Use Case Diagram

1.3.3 Task Management

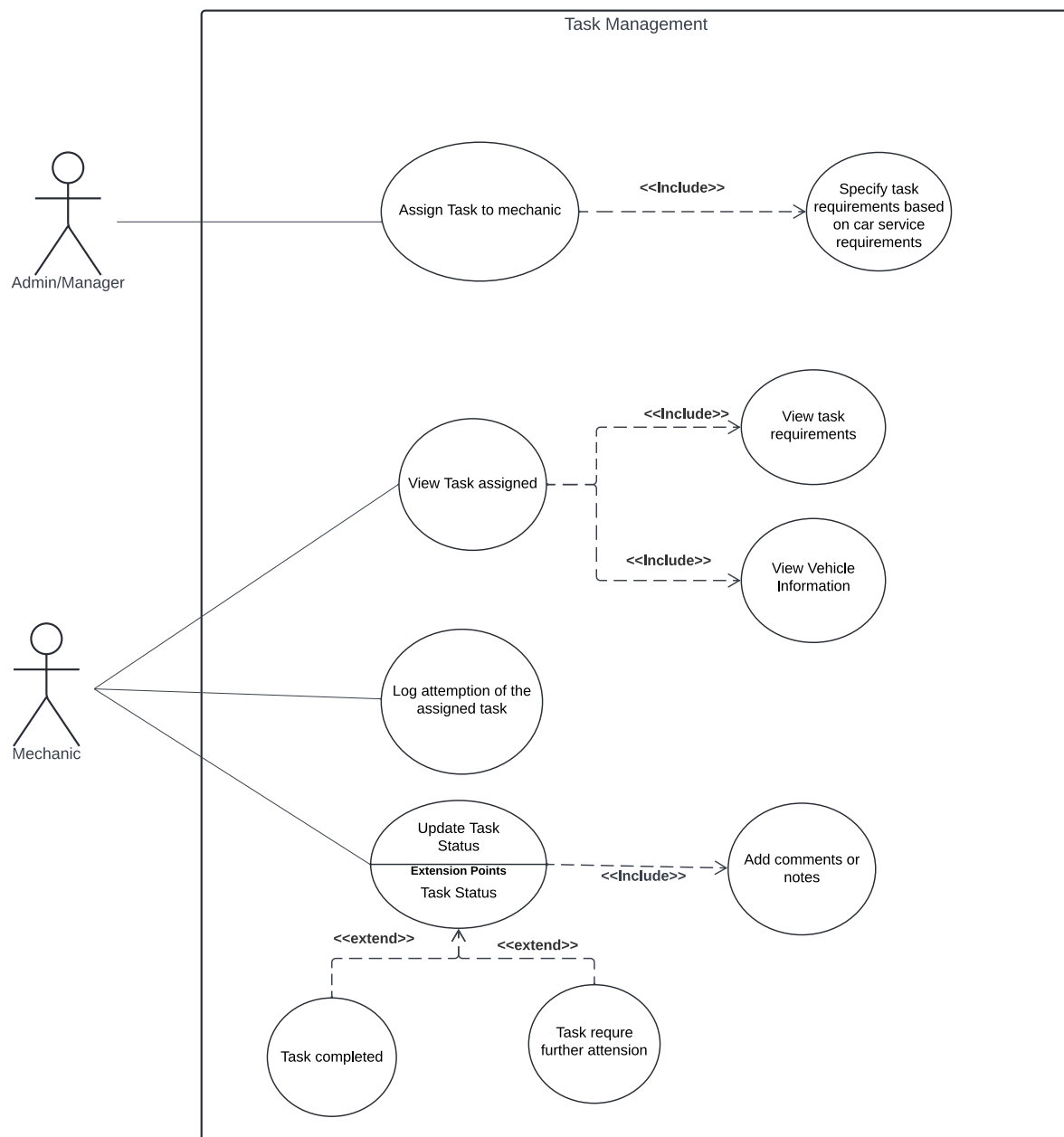


Figure 3: Figure 3: Task Management Use Case Diagram

1.3.4 Performance Monitoring

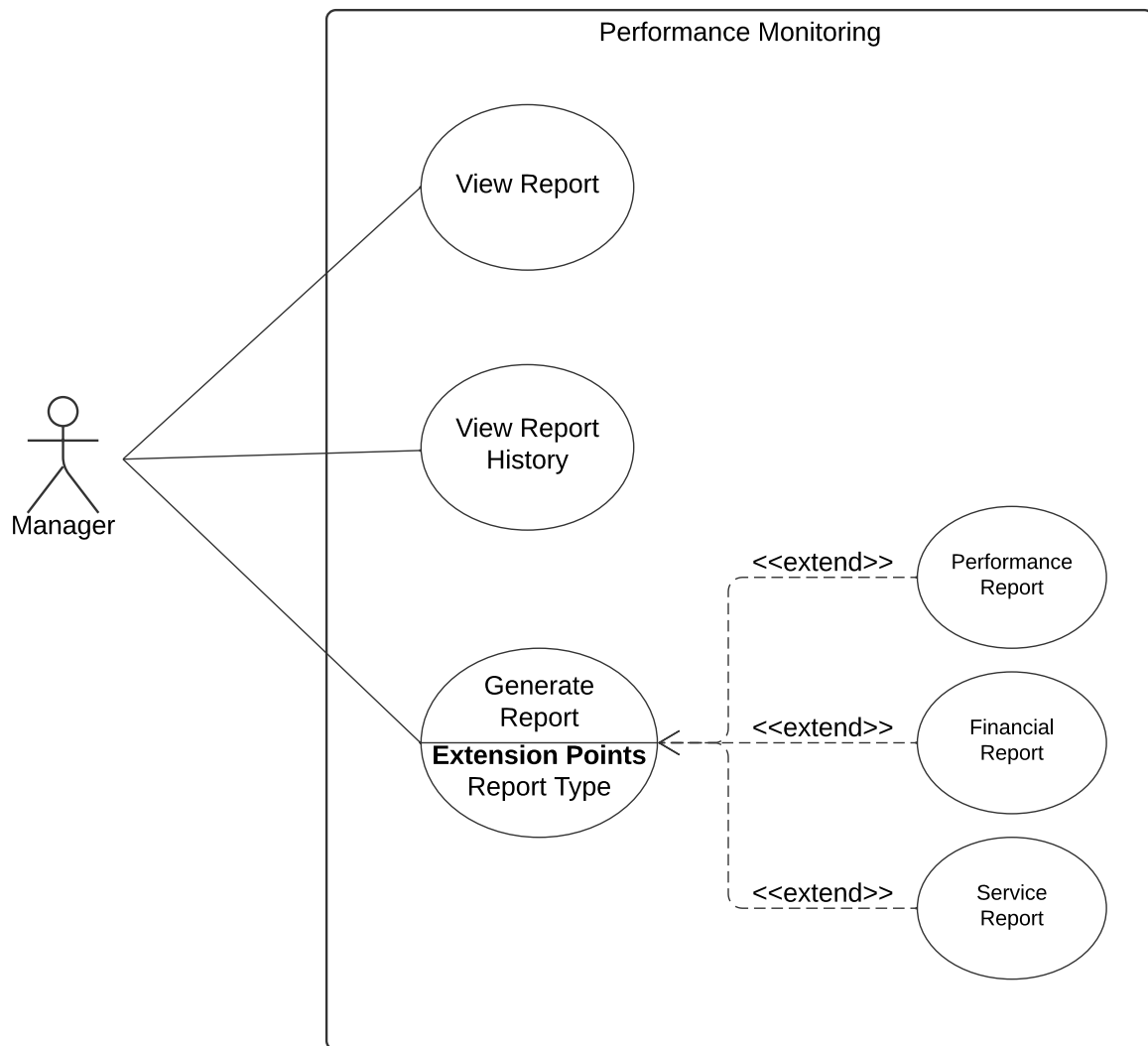


Figure 4: Skills Sharing Use Case Diagram

1.3.5 Appointment Management

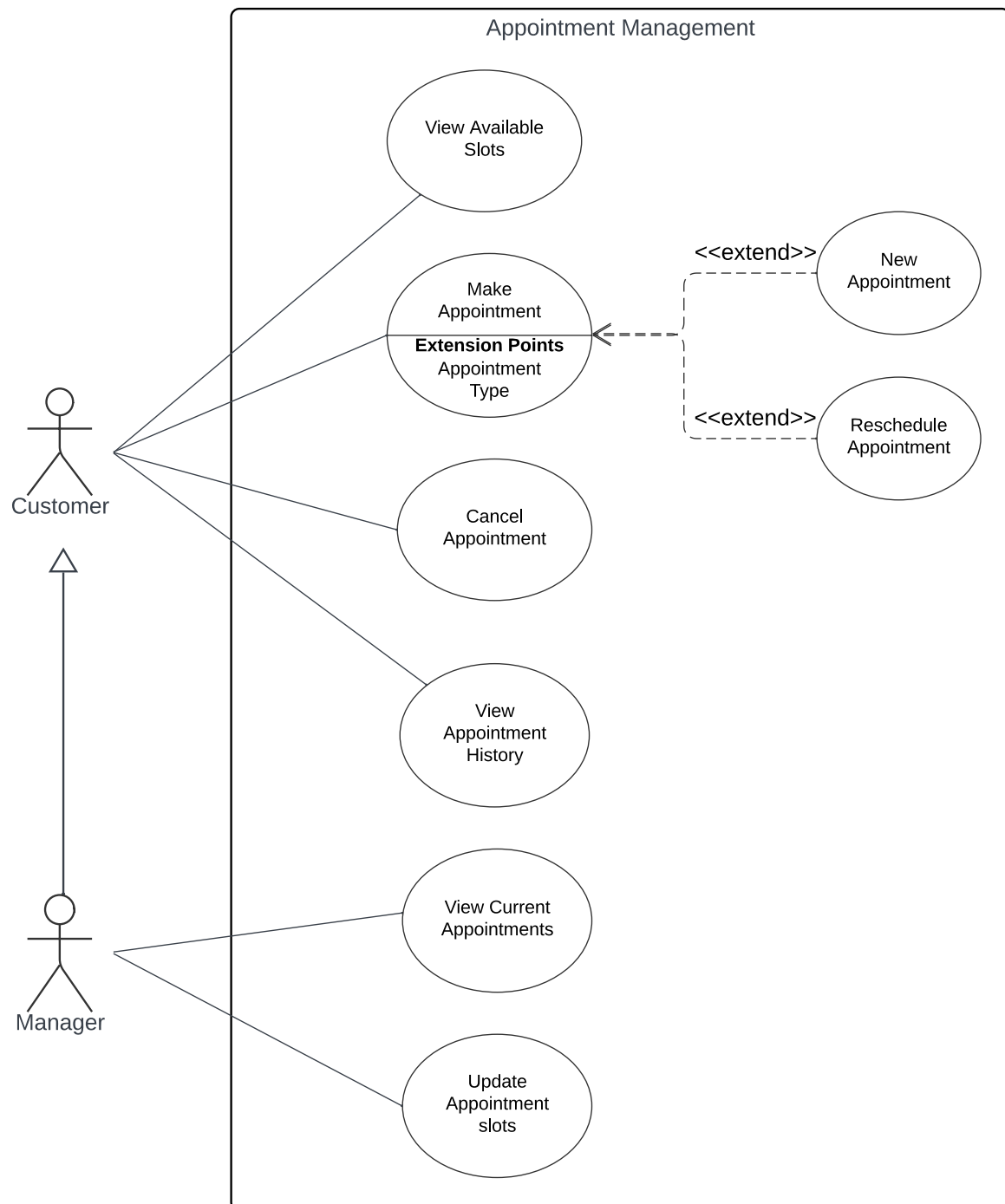


Figure 5: Appointment Management Use Case Diagram

1.3.6 Skills Sharing

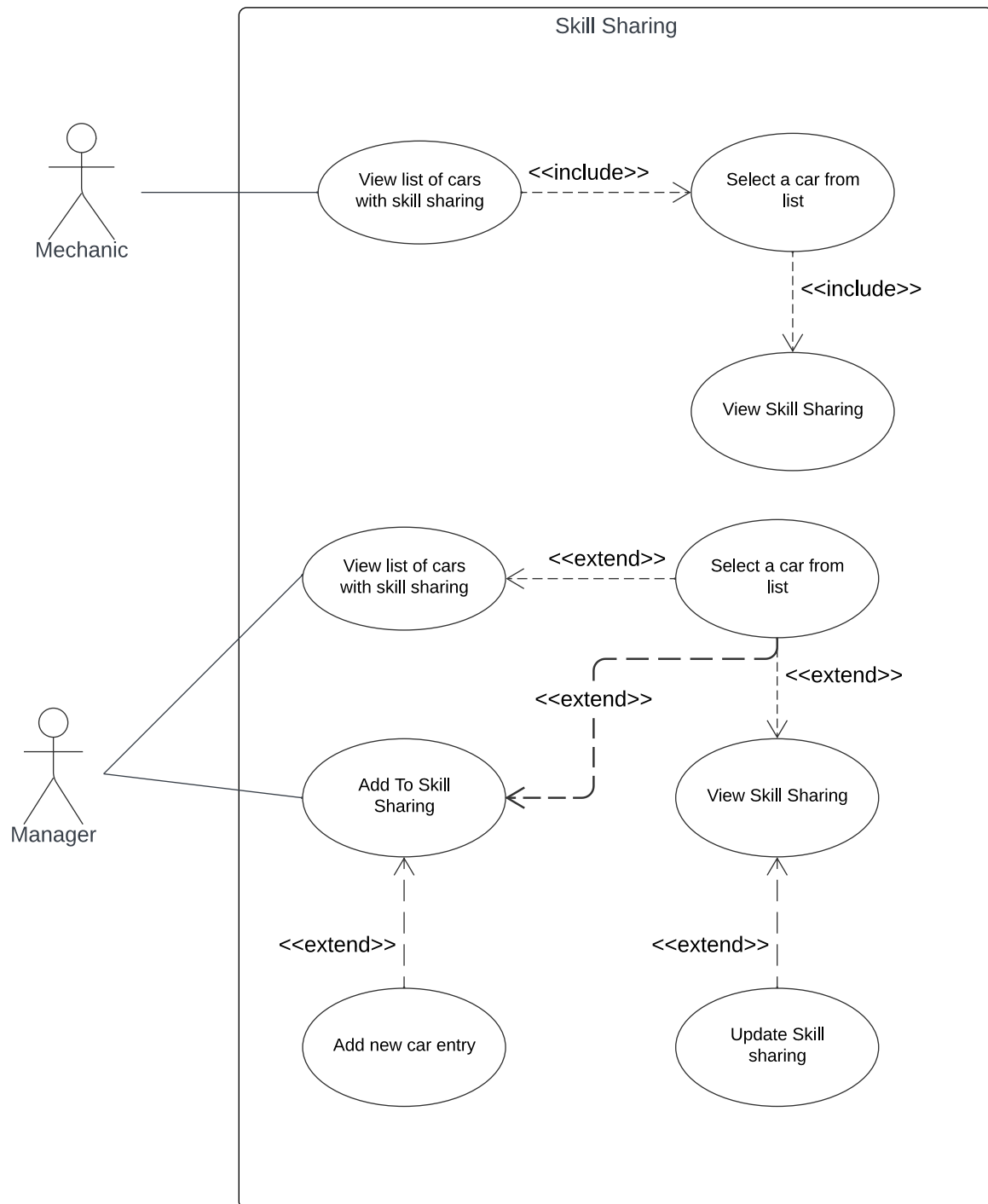
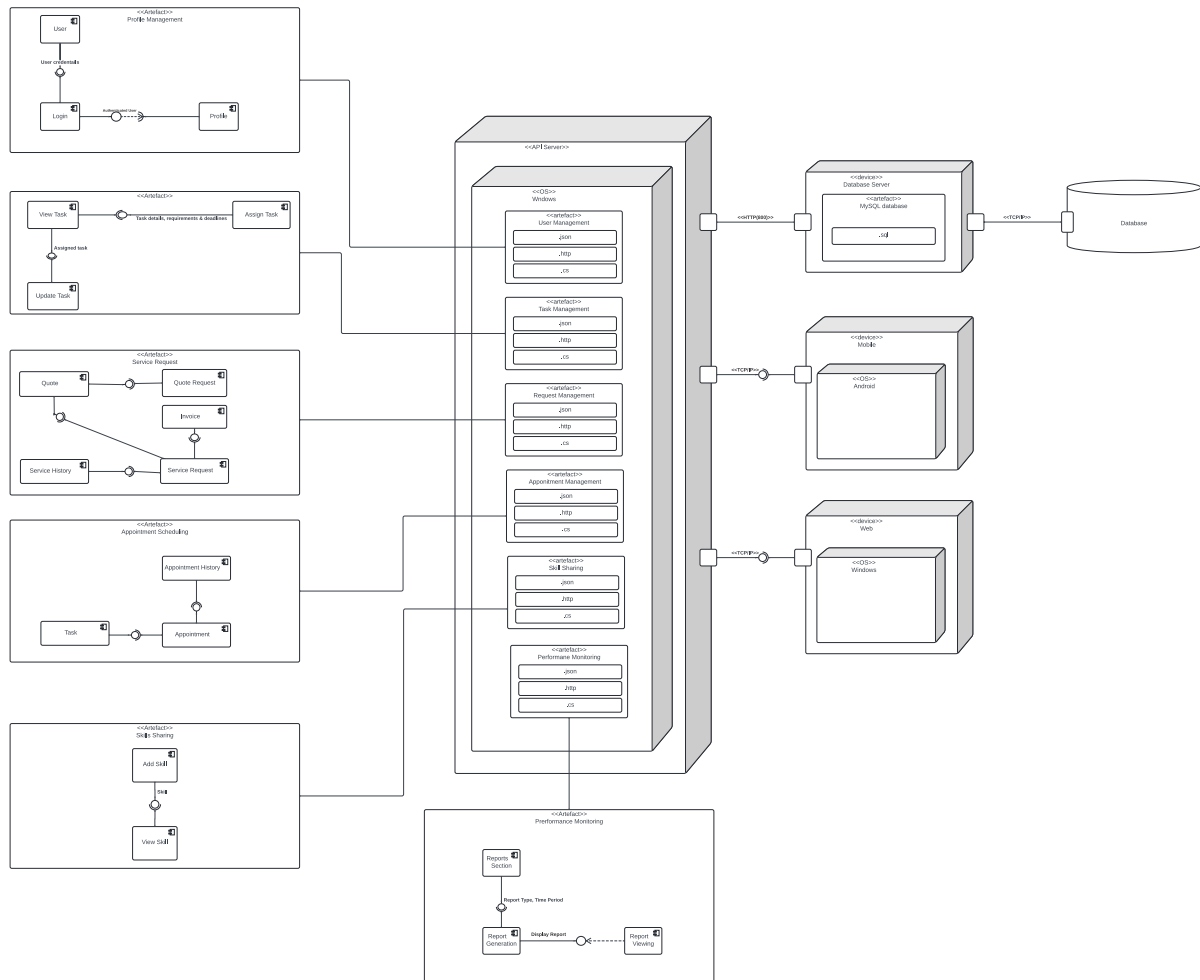


Figure 6: Skills Sharing Use Case Diagram

2. Component & Deployment Diagram



3. Comprehensive Verification and Validation (V&V) plan

3.1 User Management

#	Test Name	Input	Condition/Test	Expected Output
1.	Register	Name Surname Email address Password Date of birth Contact Number	User details correct	Register user, and continue to dashboard
			Duplicate email	Display User already exist message
2.	Login	Username Password	User credentials correct	Authenticate user, and continue to Dashboard
			User credentials incorrect	Display credentials incorrect message
3.	Reset User Credentials	Email Address Password	Credentials valid	Display reset credentials successful message
			Credentials invalid	Display Reset Credentials unsuccessful message
4.	Reset Email	Email Address	Email Address valid	Display success message
			Email Address invalid	Notify the User that the Email is invalid
			Email Address exist	Notify the User that the Email exist
5.	Reset Password	Password	Password strong	Display Password reset was successful message
			Password weak	Display password is too weak message
6.	Update profile	Names Surname Date of birth Contact Number Email	User details valid	Display personal information updated message

3.2 Service Request Management

#	Test Name	Input	Conditions/Test	Expected Output
1.	Get Service Estimation	Car Details Service Type Additional Details	Requested Service is on the System There are enough quotations on the requested service	Service Estimation. Alert Message: Estimate is subject to change.
			Requested Service is on the System There are no past quotations on the requested service.	Produce waiting time estimate for Customer. Send Quote Request to Manager.
2.	View Quote	Accept/Reject Input	Quote Accepted	Appointment scheduling window.
			Quote Rejected	Request Quote upload, to Negotiate price.
3.	Customer View Service History	Select View History	History Found	Display Service History
			History Not Found	Display Alert Message: Service History not found
4.	View Quote Request	Select View Quote Request	Available requests found	Display Available Requests
			Available requests not found	Display Alert Message: No available request found. Display Empty Table.
5.	Produce Quotation	Billing Account Client Car Details Date Parts Quantity Description	Part available Client Registered	Message: Quote send successfully
			Part not available	Message: part A is not in

				store make arrangements.
			Client not Registered	Message: Request the client to register, for appointment scheduling
6.	Assign Task	Mechanic Pending Task	Mechanic available Pending Task	Assign Task to available mechanic. Message: Task Assigned Successfully.
			Mechanic not available Pending ask	Message: Available Mechanic for the task. Set Task status to Pending. Put Task to Waiting list
			Not Pending Task	Message: No available Tasks to assign to a Mechanic
7.	Send Invoice	Billing Account Client Car Details Date Parts Quantity Description	Task Complete	Send invoice to customer
			Task not complete	Message: Task for service is not complete yet. Do not send invoice to customer
8.	Manager View Service History	Select Service History	Available Services	
			No Available Services in Service History	Message: No Available History of services

3.3 Task Management

#	Test Name	Input	Conditions/Test	Expected Output
1.	View Task	Mechanic login credentials	Mechanic logs into their dashboard on the mobile app	Access to the mobile app dashboard
		Mechanic is logged into the system	Navigate to the "Tasks" section	Access to the "Tasks" section
			View the list of assigned tasks No available tasks	List of assigned tasks with service requirements and vehicle information Message indicating no tasks currently assigned
			Click on a task to view additional details	Display of additional details for the selected task
2.	Update task	Mechanic is logged into the system	Navigate to the "Tasks" section	Access to the "Tasks" section
		Task available No tasks assigned No task selected	Select a task to update No task to select Try to update task without selecting one	Selected task confirmed Message indicating no tasks currently assigned Prompt to choose a task first
		Task selected	Update task status (in progress, completed, requiring further attention)	Task status updated
			Add comments or notes to the task update	Comments or notes added
		Task status updated	Submit the task update	Confirmation message upon successful submission
3.	Assign Task	Manager login credentials	Manager logs into their web application dashboard	Access to the web application dashboard

		Manager is logged into the system	Navigate to the "Tasks" section	Access to the "Tasks" section
			Select a task to be assigned	Selected task confirmed
		Task selected	a) Select a mechanic to assign the task to	Mechanic selected for task assignment
		Mechanic selected No mechanics available	Specify additional task details (service requirements, vehicle information, deadlines) Try to assign a task	Additional task details specified Notification prompting to assign tasks later
		Task details specified	Confirm and submit the task assignment	Confirmation message upon successful assignment

3.4 Performance Monitoring

#	Test Name	Input	Conditions/Test	Expected Output
1.	Generate report	Manager login credentials Manager is logged into the system	Navigate to the "Reports" section	Access to "Reports" section
		Report type	Select a report type (e.g., task completion rates)	Report type selection confirmed
		Time period	Select a time period/frame	Time period selection confirmed
			Generate the report	Generated report displayed or download link provided
2.	View report	Generated report with specified report type and time period	View the report	Display of the report with performance analytics

3.5 Appointment Management

#	Test Name	Input	Conditions/Test	Expected Output
1	Schedule an appointment	Customer ID, Preferred date, and time. Service requirement.	Preferred date and time exist, service required specified.	Appointment is scheduled. Display success appointment scheduling message.
2		Customer ID, Preferred date, and time. Service requirement.	Preferred date and time are not available, service required specified.	Display error message.
3	View appointment	Manager Id	User is a manager.	Display list of appointments scheduled by users.
4	Update appointment date	Manager ID, Appointment ID, Date and time	User is a manager, Appointment exists, Date and time is available	Update appointment date, display confirmation message
5		Manager ID, Appointment ID, Date and time	User is a manager, Appointment exists, Date and time is not available	Display error message, display message for manager to retry again.

3.6 Skills Sharing

#	Test Name	Input	Conditions/Test	Expected Output
1	Add Skill to Existing vehicle	Vehicle ID Skill Details	Vehicle exists, User logged in as manager	Add the skill to the vehicle records
2		Add Skill to Existing vehicle	Vehicle exist, User is not a manager	Display authentication error message
3	Add Skill to a new vehicle	Vehicle Form, Skill details	Vehicle does not exist. User is a manager	Create new vehicle and add skill to vehicle record.
4	View Skills for a vehicle	Vehicle ID	Vehicle exists, User is manager/mechanic	Display list of skills for vehicle
5		Vehicle ID	Vehicle does not exist, User id manager/mechanic	Display error message that user does not exist.
6		Vehicle ID	Vehicle exists, User is not mechanic or manager.	Display authentication error message

4. Gantt Chart & Resource Breakdown

Autoworx Barnyard Online

Team 4

Cyber Crew

Gantt Chart & Resource Breakdown

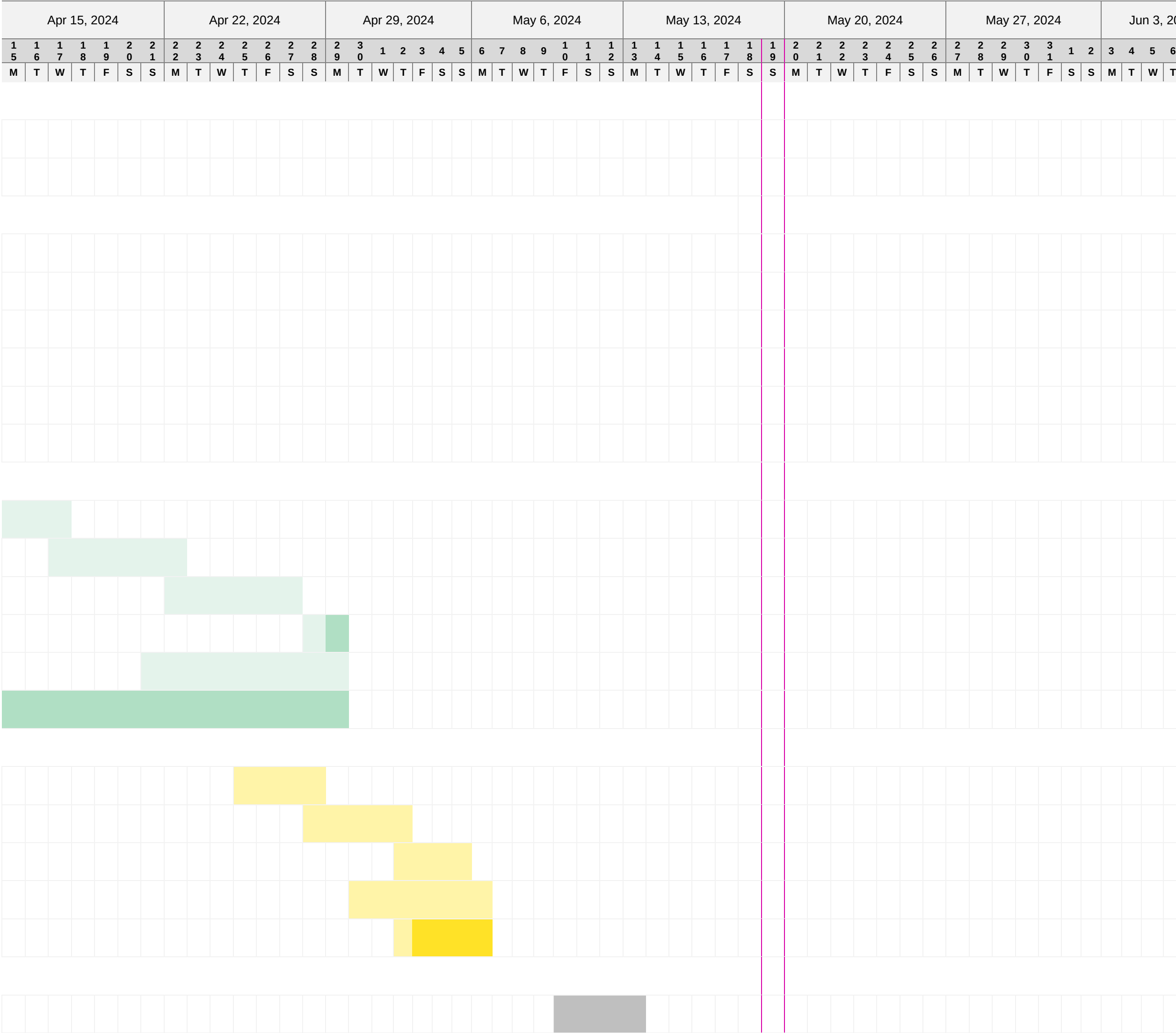
TASK	ASSIGNED TO	PROGRESS	START	END
Project Approval				
Sponsor Search	Cyber Crew	100%	2/13/24	2/16/24
Project Proposal	Cyber Crew	100%	2/16/24	3/21/24
Requirements and Analysis				
Background, Problem and Solution	Cyber Crew	100%	2/16/24	3/7/24
Requirements Extraction	Cyber Crew, Sponsor	100%	2/18/24	3/18/24
Identify and Prioritize use cases	Cyber Crew	100%	2/22/24	3/28/24
Feasibility and Risk Study	Tokelo Monare	100%	2/22/24	3/21/24
Documentation	Nkosinathi Tshabalala	100%	2/28/24	4/2/24
Problem Presentation	Cyber Crew	100%	3/5/24	4/5/24
Specification				
Updated Background, Problem and Solution	Nkosinathi Tshabalala, Tokelo MONARE	100%	4/15/24	4/17/24
Use Case Description	Mandla Ndlovu,Nkosinathi Tshabalala, Peo Vukea	100%	4/17/24	4/22/24
Activity Diagram	Cyber Crew	100%	4/22/24	4/27/24
Database Design	Peo Vukea, Tokelo Monare	95%	4/28/24	4/29/24
Documentation	Nkosinathi Tshabalala	100%	4/21/24	4/29/24
Iterative Prototype 1.0	Cyber Crew	20%	3/1/24	4/29/24
Architecture Design				
Updated Background, Problem and Solution	Nkosinathi Tshabalala	100%	4/25/24	4/28/24
Class Diagram	Cyber Crew	100%	4/28/24	5/2/24
Interaction Sequence Diagram	Cyber Crew	100%	5/2/24	5/5/24
Documentation	Nkosinathi Tshabalala	100%	4/30/24	5/6/24
Iterative Prototype 2.0	Cyber Crew	35%	5/2/24	5/6/24
Project Management				
Updated Background, Problem and Solution	Nkosithathi Tshabalala	100%	5/10/24	5/13/24

Project start:

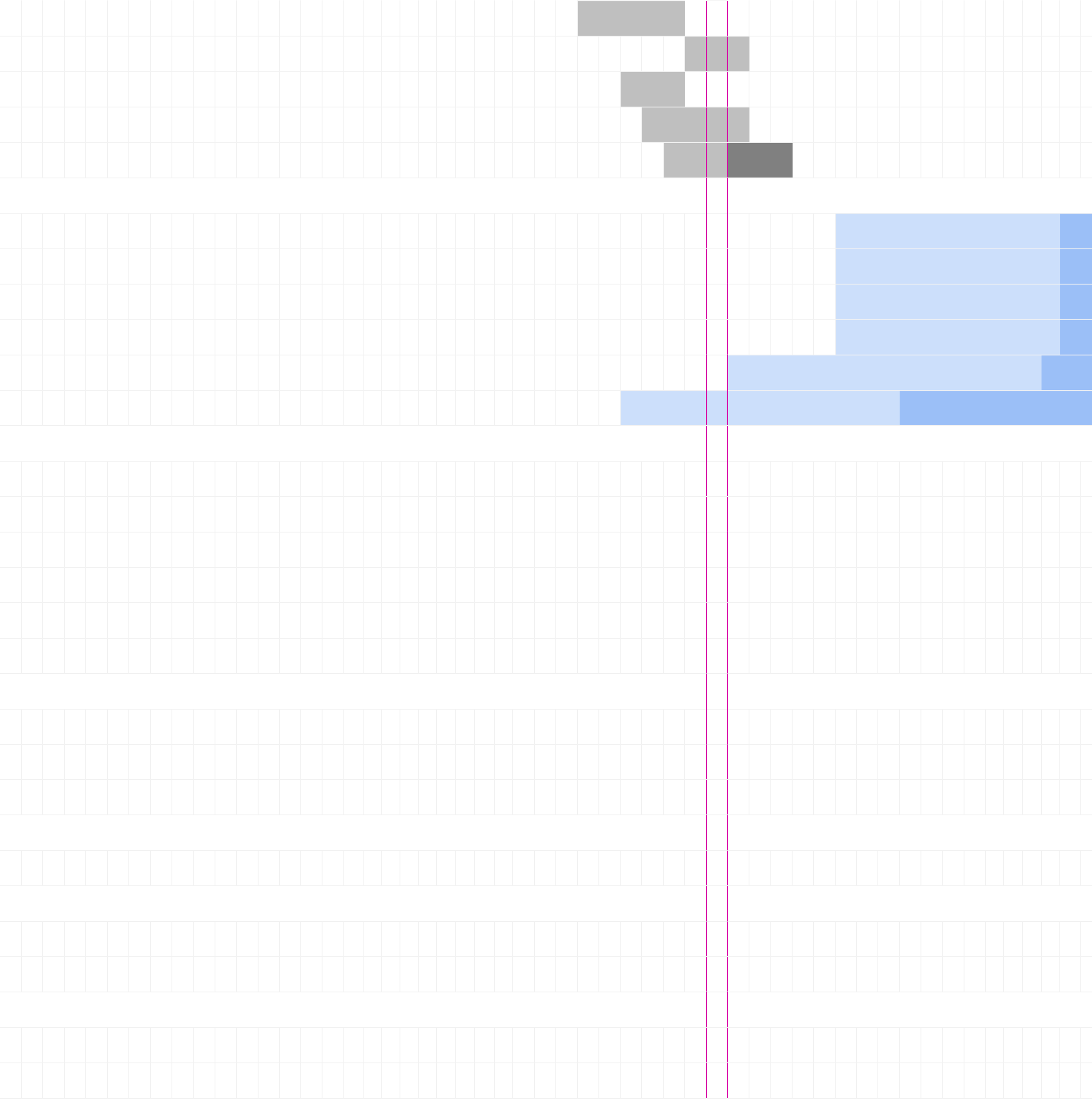
Tue, 2/13/2024

Display week:

10



Component and Deployment Diagram	Peo Vukea, Mandla Ndlovu	100%	5/13/24	5/17/24
Verification and Validation Plan	Cyber Crew	100%	5/18/24	5/20/24
Gantt Chart and Resource Breakdown	Tokelo Monare	100%	5/15/24	5/17/24
Documentation	Nkosithathi Tshabalala	100%	5/16/24	5/20/24
Iterative Prototype 3.0	Cyber Crew	65%	5/17/24	5/22/24
Alpha System				
Mobile [Xamarin]	Mandla Ndlovu	20%	5/25/24	7/19/24
Web App [Asp.net Core]	Nkosinathi Tshabalala	20%	5/25/24	7/19/24
Backend Web App[Asp.net Core]	Tokelo Monare	20%	5/25/24	7/19/24
Backend Mobile App[Asp.net Core]	Peo Vukea	20%	5/25/24	7/19/24
System and Features	Cyber Crew	25%	5/20/24	7/19/24
Structured Main Business Case	Cyber Crew	20%	5/15/24	7/19/24
Beta System				
Mobile [Xamarin]	Mandla Ndlovu	5%	7/29/24	8/28/24
Web App [Asp.net Core]	Nkosithathi Tshabalala	5%	7/29/24	8/28/24
Backend Web App[Asp.net Core]	Tokelo Monare	5%	7/29/24	8/28/24
Backend Mobile App[Asp.net Core]	Peo Vukea	5%	7/29/24	8/28/24
Complete System and Features	Cyber Crew	20%	7/29/24	8/28/24
Structured Extended Business Case	Cyber Crew	25%	8/28/24	9/1/24
Complete Documentation				
User Guide	Cyber Crew	0%	7/31/24	8/3/24
Installation Guide	Cyber Crew	0%	8/3/24	8/7/24
Developer Guide	Cyber Crew	0%	8/8/24	8/11/24
Screencast Presentation				
System Demonstration	Cyber Crew	0%	8/2/24	8/5/24
Marketing Website				
Team Marketing	Cyber Crew	0%	7/31/24	8/30/24
Project Marketing	Cyber Crew	0%	8/30/24	9/29/24
Final System				
Realistic Data in Database	Cyber Crew	0%	8/10/24	8/30/24
Final System	Cyber Crew	0%	8/30/24	10/16/24



Estimation of Total Project Cost

Resource	Total Cost
Software tools and Licences	R 550
Hardware [4x Laptops]	R 32 000
Indirect Costs	R 1000
Contingency	R 400
Labor Costs	R 0
Total Cost	R 33 950

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